

Is the Cancer Rate Generally Lower for the Japanese Population?

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The largest population in the Genetic Association Database (GAD) is Japanese (Figure 6), and therefore some disease classes reflect the highest population count as being Japanese (Figure 4). However, other disease classes reflect a lower occurrence within the Japanese population.

In particular, the Japanese population is shown as being the third highest within the database to be diagnosed with cancer (Figure 1). Thus, one may ask, is the cancer rate generally lower for the Japanese population? To begin answering this question, in Figure 2, for each disease class, the proportion of the population identifying as solely Japanese was charted along with the proportion of the total database population that is Japanese (green). Values below this population proportion are blue and those above are purple. In Figure 3 is the number of positively associated genes for cancer, commonly shared amongst the other disease classes.

Given the positive association between the plotted genes and cancer as well as other disease classes it would be reasonable to conclude that the diseases classes with lower proportions of the Japanese population, like cancer, may have commonly associated genes as well. However, many of the disease classes with the highest number of genes in common with cancer also have some of the highest proportion of Japanese people having the disease. Additionally, the disease classes with similarly lower rates as cancer also have fewer genes in common with cancer.

Therefore, the lower proportion of Japanese people within the dataset having cancer does not seem to have sufficient evidence to deem it statistically significant. Any data that can be analyzed from the dataset to support the connection between the genes positively associated with cancer and the prevalence of other disease classes within the Japanese population in fact disprove the association and the premise that cancer rates may generally be lower within the Japanese population.

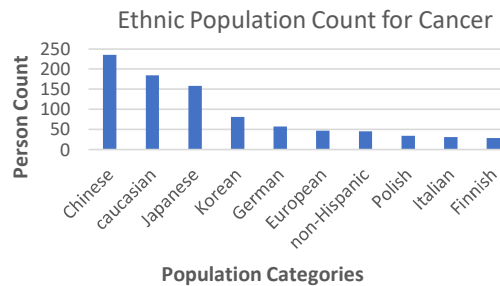


Figure 1

```
SELECT population, COUNT(population) as "num_pop", disease_class
FROM gad
WHERE population != "" and disease_class = "CANCER"
GROUP BY population
ORDER BY num_pop DESC;
```

population	num_pop	disease_class
Chinese	235	CANCER
caucasian	184	CANCER
Japanese	158	CANCER
Korean	81	CANCER
German	57	CANCER
European	47	CANCER
non-Hispanic	45	CANCER
Polish	34	CANCER
Italian	31	CANCER
Finnish	29	CANCER
African Ame...	29	CANCER
Indian	26	CANCER
Tunisian	23	CANCER

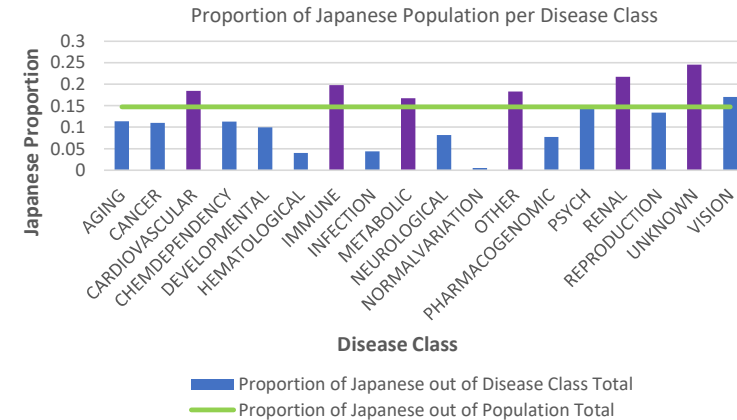


Figure 2

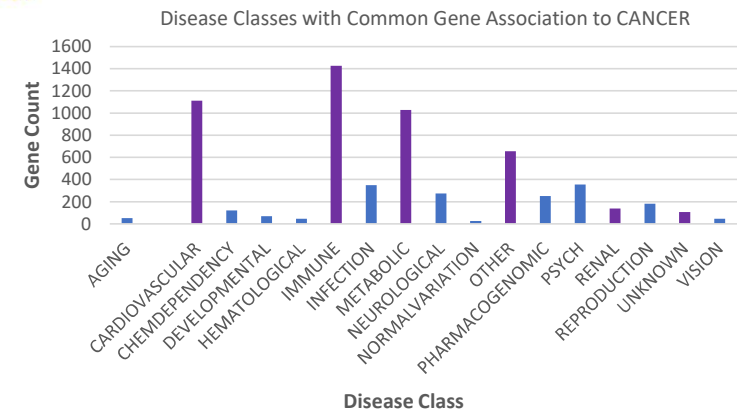


Figure 3



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Becker *et al.* (2004). The Genetic Association Database. *Nature Genetics* 36(5):431-2

```

SELECT population, COUNT(population) as "num_pop", disease_class
FROM gad
WHERE population != "" and disease_class IN (SELECT disease_class
FROM (SELECT disease_class, COUNT(disease_class) as "num_disease"
FROM gad
WHERE population != "" and disease_class != ""
GROUP BY disease_class
ORDER BY num_disease DESC
LIMIT 5) as top5)
GROUP BY population, disease_class
ORDER BY num_pop DESC;

```

Figure 4

population	num_pop	disease_class
Japanese	478	IMMUNE
Japanese	291	METABOLIC
Chinese	235	CANCER
Chinese	225	METABOLIC
Japanese	211	CARDIOVASCULAR
Chinese	206	IMMUNE
Caucasian	203	IMMUNE
Caucasian	201	METABOLIC
Chinese	190	CARDIOVASCULAR
caucasian	184	CANCER
Japanese	158	CANCER
Korean	141	IMMUNE
German	125	IMMUNE

```

SELECT population, COUNT(disease_class) as "num_disease", disease_class
FROM gad
WHERE population = "Japanese" and disease_class != ""
GROUP BY disease_class
ORDER BY num_disease DESC;

```

Figure 5

population	num_disease	disease_class
Japanese	478	IMMUNE
Japanese	291	METABOLIC
Japanese	211	CARDIOVASCULAR
Japanese	158	CANCER
Japanese	105	PSYCH
Japanese	77	OTHER
Japanese	56	NEUROLOGICAL
Japanese	45	UNKNOWN
Japanese	30	CHEMDEPENDENCY
Japanese	28	RENAL
Japanese	23	VISION
Japanese	23	REPRODUCTION
Japanese	19	INFECTION

More Charts and Queries

```

SELECT disease_class, COUNT(disease_class) as "count"
FROM (SELECT disease_class, gene, association, gene_name
FROM gad
WHERE disease_class != "CANCER") as temp
WHERE association = "Y" AND gene IN (
SELECT gene
FROM gad
WHERE disease_class = "CANCER")
GROUP BY disease_class;

```

Figure 8

disease_class	count
NEUROLOGICAL	274
PSYCH	356
OTHER	654
IMMUNE	1428
UNKNOWN	106
PHARMACOGENOMIC	251
INFECTION	350
RENAL	137
CARDIOVASCULAR	1113
METABOLIC	1029
AGING	51
CHEMDEPENDENCY	122
NORMALVARIATION	26

```

SELECT COUNT(disease_class) as "num_disease", disease_class
FROM gad
WHERE population != "" and disease_class != ""
GROUP BY disease_class
ORDER BY num_disease DESC;

```

Figure 7

num_disease	disease_class
2418	IMMUNE
1741	METABOLIC
1436	CANCER
1142	CARDIOVASCULAR
701	PSYCH
685	NEUROLOGICAL
434	INFECTION
420	OTHER
375	NORMALVARIATION
266	CHEMDEPENDENCY
208	PHARMACOGENOMIC
183	UNKNOWN
172	REPRODUCTION

```

SELECT population, COUNT(population) as "num_pop"
FROM gad
WHERE population != ""
GROUP BY population
ORDER BY num_pop DESC;

```

Figure 6

population	num_pop
Japanese	1589
Chinese	1317
Caucasian	1068
Korean	597
Italian	360
European	315
German	295
Indian	269
Spanish	231
Finnish	219
Turkish	166
French	153
African A...	152

